

課題について :eax レジスタの値を求めよ processor 32 bit である

Command	action	eax	ebx	ecx
0x8048628				
Mov eax, 0x9	eax= 0x9	0x00000009	-	-
Lea ebx, [eax+0x3]	Ebx =eax +3=12	0x00000009	0x0000000C	-
Shl al, 0x2	al=al<<2	0x00000024	0x0000000C	-
Xor bl, al	bl = bl XOR al	0x00000024	0x00000028	-
Mov ecx, ebx	ecx = ebx	0x00000024	0x00000028	0x00000028
Sub ecx, eax	ecx =ecx-eax	0x00000024	0x00000028	0x00000004
Dec ecx	ecx = eax-1	0x00000024	0x00000028	0x00000003
Imul eax, ecx	eax = eax*ecx	0x0000006c	0x00000028	0x00000003

- 変更なし

1. mov eax, 0x9

eax: 0x00000009 (9) ebx と ecx 変更なし

2.lea ebx, (eax+0x3), eax: 0x00000009 (9) , ebx: $eax + 3 = 9 + 3 = 12$ - 0x0000000C

3. shl al, 0x2 eax:

first 0x00000009 AL = 0x09 (00001001₂) - 2 binary

AL << 2 00100100₂ = 0x24 – 16 binary xor bl, al. {eax: 0x00000024 (36)}

4. ebx:

first 0x0000000C BL = 0x0C (00001100₂)

BL XOR AL: AL = 0x24 – 16binary (00100100)₂

BL = 0x0C 00001100₂

XOR: 00100100₂ ⊕ 00001100₂ = 00101000₂ = 0x28(16 binary)

5. mov ecx, ebx

eax: 0x00000024 (36)

ebx: 0x00000028 (40)

ecx , ebx- 0x00000028 (40)

6. Subtract: $ecx - eax = 40 - 36 = 4$; $0x28 - 0x24 = 0x04$. {0x00000004 (4)}

7. Initial: 0x00000004 (4) Decrement: $4 - 1 = 3$ 0x00000003 (3)

8. imul eax, ecx

Initial: 0x00000024 (36)

Multiplication: $eax * ecx = 36 * 3 = 108$ (10 binary) 0x6C (16 binary)

Result:

eax: 0x0000006C (16 binary) 108 (10 binary)

ebx : 0x00000028 (16 binary) 40 (10 binary)

ecx : 0x00000003 (16 binary) 3 (10 binary)